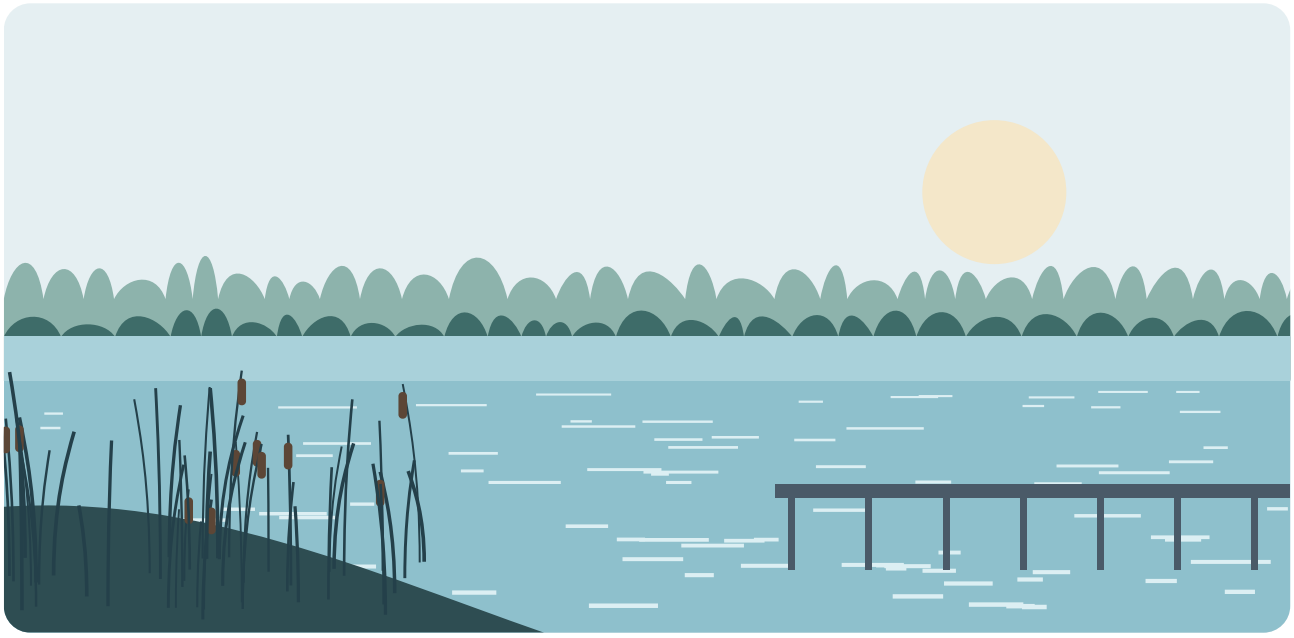
Metro Lakes Watershed District

Monitoring and protecting the lakes, ponds and streams of our member cities

2025 Water Quality Report

Chloride, total phosphorus and water clarity at district lakes, inlets and outlets

Annual monitoring report to the Board of Managers | Revised edition, October 8, 2026



REVISED EDITION, OCTOBER 8, 2026

This edition replaces the report posted June 11, 2026. Chloride results for three Phase 1 lake sites, COR-1 (Corliss Lake), PEN-1 (Penner Lake) and ODG-1 (Lake Odegaard), have been corrected. See the errata on page 3.

Prepared by the district's water resources program
metrolakeswd.org/reports/2025-water-quality

Contents

Errata: revised edition, October 8, 2026 3
1. Monitoring program 4
2. Winter salt use in member cities 5
3. Chloride 6
Table 3. Annual mean chloride by monitoring site, 2020 and 2025 7
4. Phosphorus and water clarity 8
Appendix A. Phase 3 inlet and outlet stations 9
Appendix B. Quality assurance and data availability 10

2025 at a glance

- The district monitored chloride at 10 sites in 2025: eight open-water lake sites, plus a storm sewer inlet and a lake outlet added this year as Phase 3.
- Annual mean chloride was above the state's chronic standard of 230 mg/L at two lake sites, Lake Odegaard (ODG-1) and Haskell Pond (HSK-1).
- All five Phase 1 lakes had higher annual mean chloride in 2025 than in 2020, the program's first full year of monitoring.
- During March snowmelt, chloride at the Lake Odegaard north inlet reached 1,020 mg/L, several times the lake's annual mean.
- Member cities reported using 9% more road salt in the 2024-25 winter than in the mild 2023-24 winter. Participants in the district's Smart Salting training reported using 22% less.
- Summer phosphorus and water clarity met state standards at six of the eight lake sites.

About this revised edition

The edition posted June 11, 2026, overstated 2025 chloride at three Phase 1 lake sites because of a dilution-factor error at the district's contract laboratory. That edition listed four lake sites above the chronic standard; this edition lists two. Please use the revised figures. The errata on page 3 list every change.

Errata: revised edition, October 8, 2026

This revised edition of the 2025 Water Quality Report replaces the edition posted on June 11, 2026. Please use the revised figures in any further use of the report.

What happened

A dilution-factor error in contract-lab batch 25-0217 (samples collected Feb. 3 to April 28, 2025) inflated open-water composite results for COR-1, PEN-1 and ODG-1.

The laboratory reported that batch with an incorrect dilution factor. It recalculated the results from its original instrument records using the correct factor and reissued its report for the batch. No samples were reanalyzed. Samples from the other monitoring sites collected in the same period were analyzed in other batches and are not affected.

How the error was found

After the June edition was posted, staff reviewing the first year of Phase 3 data found that the 2025 mean at the Penner Lake outlet (PEN-OUT, 201 mg/L) was lower than the published open-water mean for Penner Lake (241 mg/L). The outlet carries water from the lake's surface, so the two should agree closely. A review of laboratory records traced the difference to batch 25-0217.

Corrected 2025 annual means

Site	Water body	2025 annual mean, June 11 edition (mg/L)	Revised 2025 annual mean (mg/L)	Difference (mg/L)	Chronic standard, June 11 edition	Chronic standard, revised
COR-1	Corliss Lake, boat launch	268	214	-54	Above	Below
PEN-1	Penner Lake	241	198	-43	Above	Below
ODG-1	Lake Odegaard	312	247	-65	Above	Above

Values are 2025 annual means in milligrams per liter (mg/L). The 2020 annual means in Table 3 are not affected.

What changed in the report

- Lake sites above the chronic standard in 2025: the June edition listed four (COR-1, PEN-1, ODG-1 and HSK-1). This edition lists two (ODG-1 and HSK-1).
- Revised: the summary (page 2), the Chapter 3 results and Figure 2 (page 6), Table 3 (page 7) and the outlet comparison in Appendix A (page 9). A data review note has been added to Appendix B (page 10).
- Pages from Chapter 1 onward are numbered one higher than in the June edition because this errata page was added.

What did not change

- 2025 annual means for the other seven monitoring sites: ARD-1, HSK-1, SUM-1, KET-1, LTM-1, ODG-IN and PEN-OUT.
- All 2020 values, the winter salt survey in Chapter 2, and the phosphorus and water clarity results in Chapter 4.

The corrections are also posted as a separate errata table at metrolakeswd.org/reports/2025-errata. Questions: Luis Ferreira (he/him), Communications Coordinator, (651) 555-0158, luis.ferreira@metrolakeswd.org.

1. Monitoring program

The Metro Lakes Watershed District monitors water quality in the lakes, ponds and streams of its member cities. This report covers the 2025 calendar year.

Chloride monitoring has grown in three phases. Phase 1 began in 2019 at five lakes. Phase 2 added three lakes in 2022. Phase 3, new in 2025, added two stations that measure chloride moving into and out of lakes: a storm sewer inlet on the north side of Lake Odegaard and the outlet culvert of Penner Lake. Phase 1 monitoring began partway through 2019, so 2020 is the first full year of record and the comparison year used in this report.

Table 1. Chloride monitoring sites, 2025

Site ID	Water body or station	Station type	Program phase (first year)	Latitude	Longitude
ARD-1	Lake Ardell	Open water	Phase 1 (2019)	45.0187	-93.1642
COR-1	Corliss Lake, boat launch	Open water	Phase 1 (2019)	45.0412	-93.1287
PEN-1	Penner Lake	Open water	Phase 1 (2019)	45.0563	-93.1519
ODG-1	Lake Odegaard	Open water	Phase 1 (2019)	45.0734	-93.1196
HSK-1	Haskell Pond	Open water	Phase 1 (2019)	45.0291	-93.1083
SUM-1	Sumac Lake, fishing pier	Open water	Phase 2 (2022)	45.0645	-93.1768
KET-1	Kettle Lake	Open water	Phase 2 (2022)	45.0886	-93.1405
LTM-1	Little Tamarack Lake	Open water	Phase 2 (2022)	45.0953	-93.1652
ODG-IN	Lake Odegaard north inlet (storm sewer)	Inlet	Phase 3 (2025)	45.0792	-93.1224
PEN-OUT	Penner Lake outlet culvert	Outlet	Phase 3 (2025)	45.0528	-93.1461

Coordinates are decimal degrees (WGS 84). Figure 1 on the next page shows the sites, and an interactive station map is at metrolakeswd.org/monitoring/stations.

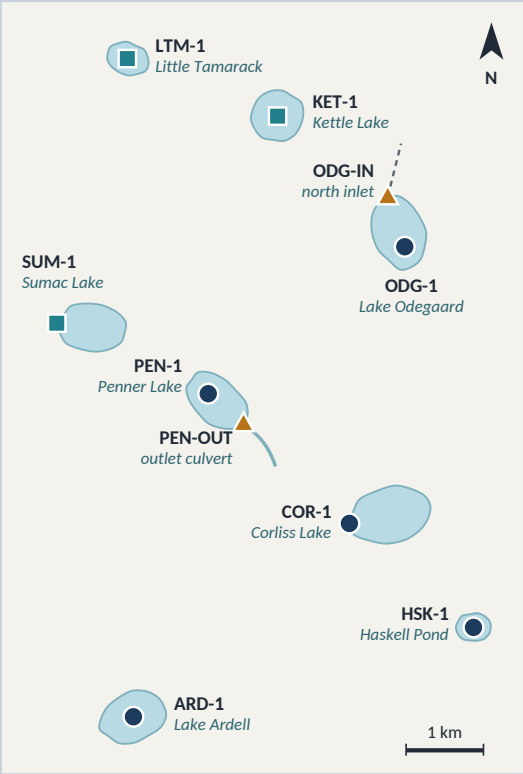
How the district samples

Sampling. Open-water lake sites are sampled every four weeks through the year, including under the ice when conditions are safe. At each site, staff combine grab samples taken at the station on the sampling date into one open-water composite sample. Phase 3 stations are sampled every four weeks while water is flowing, with added samples during snowmelt and after storms. A district flow sensor at the Lake Odegaard inlet records flow continuously.

Laboratory analysis. Chloride is analyzed by a state-accredited contract laboratory using EPA Method 300.0 (ion chromatography). The same laboratory analyzes total phosphorus. Water clarity is measured in the field with a Secchi disk at open-water sites from June through September.

Annual means. The annual mean for a site is the average of all of that site's results for the calendar year. Summer means for phosphorus and clarity use the June through September results.

Figure 1. Chloride monitoring sites by program phase



Program phase

- Phase 1 (2019): five open-water lake sites
- Phase 2 (2022): three open-water lake sites
- Phase 3 (2025): a storm sewer inlet and a lake outlet

Map features

- Lake (outline is schematic)
- Lake outlet channel
- Storm sewer

Sites are plotted from the coordinates in Table 1. Site codes: ARD Lake Ardell, COR Corliss Lake, PEN Penner Lake, ODG Lake Odegaard, HSK Haskell Pond, SUM Sumac Lake, KET Kettle Lake, LTM Little Tamarack Lake. IN marks an inlet station and OUT an outlet station.

2. Winter salt use in member cities

Each spring the district asks member cities how much road salt their street crews applied the winter before.

- Combined, member cities reported using 9% more road salt in the 2024-25 winter than in the 2023-24 winter.
- The 2023-24 winter was unusually mild, with little snow, so the comparison year was a low-use year. Cities that reported increases most often cited more frequent snow and ice events in 2024-25.
- The survey covers city streets only. County and state roads, school districts and private applicators who treat parking lots, driveways and sidewalks are not included.

Smart Salting training

Participants who completed the district's Smart Salting training reported cutting their salt use by 22% compared with the previous winter. The Smart Salting Summary 2025 has the details, the 2026 workshop schedule and the district's cost-share program for spreader calibration equipment (metrolakeswd.org/smartsalting).

Why winter salt matters for lakes

Chloride does not break down or settle out. Once it reaches a lake, it leaves only with the water that flows out through the outlet. Some salt also soaks into soil and shallow groundwater, which can carry it to lakes in other seasons. Lowering chloride in district lakes depends on putting less salt down in the first place.

3. Chloride

Chloride reaches district lakes mainly from road salt and other deicers. At high concentrations it harms fish, insects and other aquatic life. Minnesota's water quality standards set two limits for chloride in lakes and streams (Table 2).

Table 2. Minnesota chloride standards for aquatic life

Standard	Concentration	Protects against	How this report uses it
Chronic standard	230 mg/L	Sustained exposure	Compared with each lake site's annual mean as a screening check
Maximum (acute) standard	860 mg/L	Short exposures	Reference for short-term peaks, such as snowmelt at inlets

Formal impairment assessments are made by the Minnesota Pollution Control Agency using its own methods. Comparisons in this report are for screening.

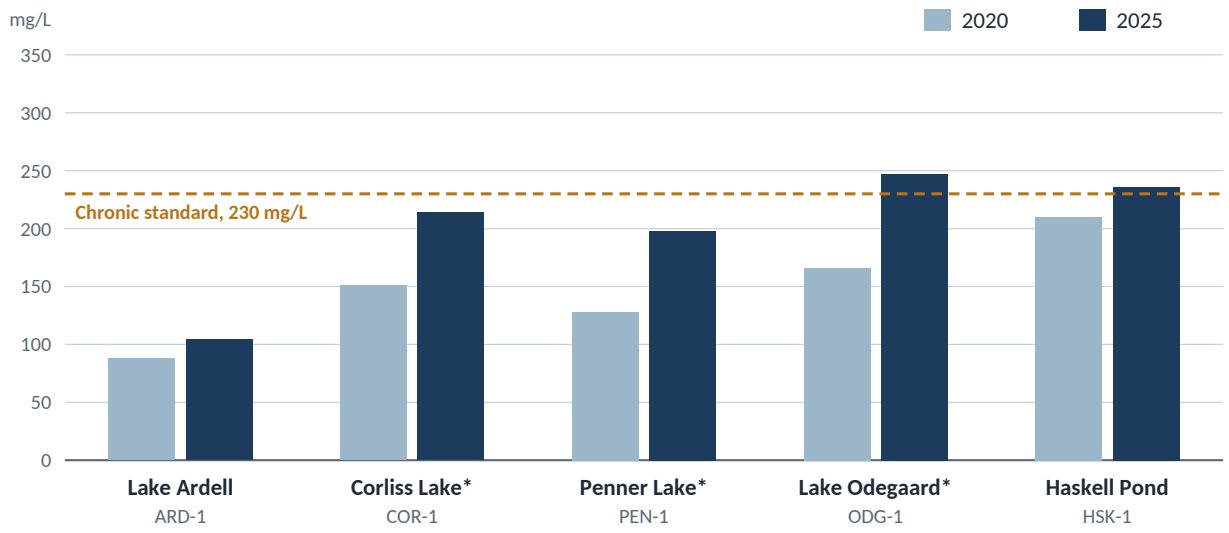
2025 results

- Two lake sites had 2025 annual means above the chronic standard: Lake Odegaard (ODG-1, 247 mg/L) and Haskell Pond (HSK-1, 236 mg/L).
- Corliss Lake (COR-1) and Penner Lake (PEN-1) were below the standard in 2025. The June 11 edition reported both above it; see the errata on page 3.
- Lake Ardell (ARD-1) had the lowest annual mean of the Phase 1 lakes. All three Phase 2 lakes were below the standard.
- Site-by-site annual means for 2020 and 2025 are in Table 3 on page 7.

Change since 2020

All five Phase 1 lakes had higher annual means in 2025 than in 2020 (Figure 2). Two years show the size of a change but not a trend: one year's mean also reflects that winter's snow, the salt put down and how quickly each lake flushes. The district plans a trend analysis of the full Phase 1 record in a future report.

Figure 2. Annual mean chloride at Phase 1 lake sites, 2020 and 2025. The dashed line marks the chronic standard. * 2025 value revised October 8, 2026 (see errata, page 3).



3. Chloride (continued)

Table 3. Annual mean chloride by monitoring site, 2020 and 2025 (mg/L)

METRO LAKES WATERSHED DISTRICT MONITORING DATABASE

Table 3. Annual mean chloride by monitoring site, 2020 and 2025

Milligrams per liter (mg/L). Calendar-year means of all samples at each site.

Site ID	Water body or station	Station type	Program phase (first year)	2020 annual mean	2025 annual mean	Above 230 mg/L in 2025?
ARD-1	Lake Ardell	Open water	Phase 1 (2019)	88	104	No
COR-1	Corliss Lake, boat launch	Open water	Phase 1 (2019)	151	214*	No*
PEN-1	Penner Lake	Open water	Phase 1 (2019)	128	198*	No*
ODG-1	Lake Odegaard	Open water	Phase 1 (2019)	166	247*	Yes
HSK-1	Haskell Pond	Open water	Phase 1 (2019)	210	236	Yes
SUM-1	Sumac Lake, fishing pier	Open water	Phase 2 (2022)	--	121	No
KET-1	Kettle Lake	Open water	Phase 2 (2022)	--	143	No
LTM-1	Little Tamarack Lake	Open water	Phase 2 (2022)	--	61	No
ODG-IN	Lake Odegaard north inlet (storm sewer)	Inlet	Phase 3 (2025)	--	388 (a)	n/a (b)
PEN-OUT	Penner Lake outlet culvert	Outlet	Phase 3 (2025)	--	201	n/a (b)

-- Not monitored in 2020. Phase 2 sites began in 2022 and Phase 3 stations in 2025.

* Revised 10/08/2026. Corrected for a dilution-factor error in contract-lab batch 25-0217 (samples collected Feb. 3 to Apr. 28, 2025). See errata, p. 3.

(a) Mean of 2025 samples. Highest result 1,020 mg/L in March 2025, during snowmelt. See Appendix A.

(b) Inlet and outlet stations are not lake sites and are not compared with the lake standard.

Minnesota chronic chloride standard for aquatic life: 230 mg/L. Lake sites are compared using the annual mean as a screening check.

REVISED
OCT 08 2026 MLWD WRP

Table 3 is reproduced as a scanned page from the district's monitoring database. Values are calendar-year means in milligrams per liter. Revised October 8, 2026: values for COR-1, PEN-1 and ODG-1 are corrected (see errata, page 3). Site-level data are available in spreadsheet form on request at metrolakeswd.org/data-request.

4. Phosphorus and water clarity

Phosphorus is the nutrient that most often limits algae growth in Minnesota lakes. Water clarity, measured as the depth at which a Secchi disk disappears from view, shows how far light reaches into the water. Minnesota's lake eutrophication standards for this region differ for deep and shallow lakes. Results are summer means, June through September 2025.

Table 4. Summer total phosphorus and water clarity at lake sites, 2025

Site	Lake	Lake type	Summer mean total phosphorus (µg/L)	Summer mean Secchi depth (m)	Meets state standards?
ARD-1	Lake Ardell	Deep	31	2.3	Yes
COR-1	Corliss Lake	Deep	46	1.3	No
PEN-1	Penner Lake	Shallow	52	1.2	Yes
ODG-1	Lake Odegaard	Deep	38	1.6	Yes
HSK-1	Haskell Pond	Shallow	91	0.7	No
SUM-1	Sumac Lake	Deep	27	2.9	Yes
KET-1	Kettle Lake	Deep	35	2.0	Yes
LTM-1	Little Tamarack Lake	Shallow	44	1.5	Yes

Standards for the North Central Hardwood Forests ecoregion: deep lakes, total phosphorus 40 µg/L or less and Secchi depth 1.4 m or more; shallow lakes, total phosphorus 60 µg/L or less and Secchi depth 1.0 m or more. The state standards also include chlorophyll-a, which is not shown here. µg/L: micrograms per liter. m: meters.

Results

Six of the eight lake sites met both standards. Corliss Lake did not meet the deep-lake standards for phosphorus or clarity. Haskell Pond, a shallow pond, had the highest phosphorus and the lowest clarity of the eight lakes. Phosphorus and clarity are measured separately from chloride, and the correction described on page 3 does not affect them.

Phosphorus results for the two Phase 3 stations are in Appendix A.

Appendix A. Phase 3 inlet and outlet stations

Phase 3 began in 2025 with two stations that show how chloride moves through the lake system. The Lake Odegaard north inlet (ODG-IN) is a storm sewer outfall that drains streets north of the lake. The Penner Lake outlet culvert (PEN-OUT) carries water leaving Penner Lake.

Table A-1. Phase 3 stations, 2025

Station	Description	Station type	2025 mean chloride (mg/L)	2025 mean total phosphorus (µg/L)	Notes
ODG-IN	Lake Odegaard north inlet (storm sewer outfall)	Inlet	388	184	Highest chloride result 1,020 mg/L, in March, during snowmelt
PEN-OUT	Penner Lake outlet culvert	Outlet	201	48	Sampled while water is flowing

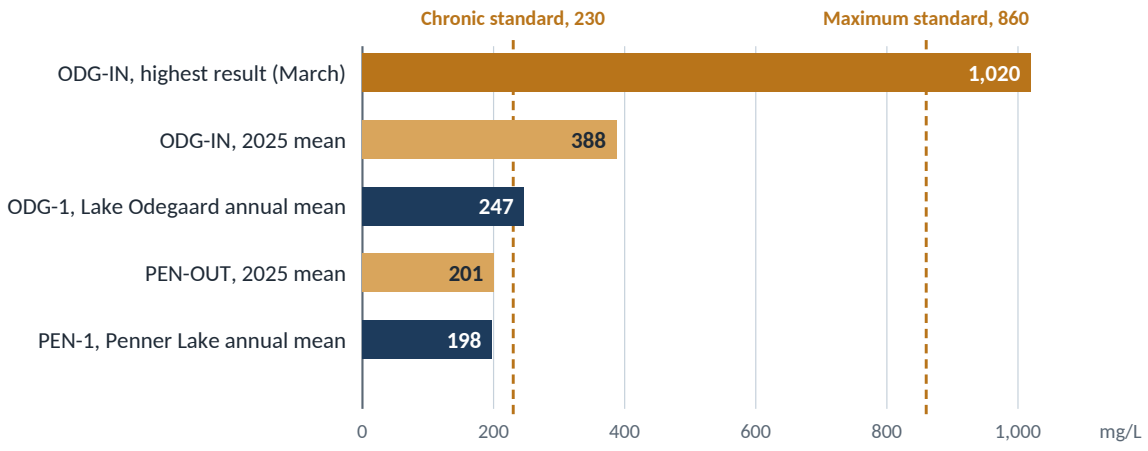
Snowmelt pulses at the inlet

Chloride at the Lake Odegaard inlet rose sharply during March snowmelt, when meltwater carried salt off streets and parking lots. The highest result, 1,020 mg/L, was several times the lake's annual mean of 247 mg/L. Inlet results are not compared with lake standards, because the inlet is a storm sewer and not a lake, but they show when chloride enters the lake and in what concentrations.

Outlet compared with the lake

The 2025 mean at the Penner Lake outlet (201 mg/L) is close to the revised open-water mean for Penner Lake (198 mg/L), as expected for a station that carries water from the lake's surface. The same comparison flagged the batch 25-0217 error in the June edition (see page 3).

Figure A-1. 2025 chloride at the Phase 3 stations and the lakes they feed or drain (mg/L). Dashed lines mark the chronic and maximum standards.



Appendix B. Quality assurance and data availability

Quality assurance

- Field staff collect duplicate samples and field blanks during the year and keep field notes for every visit.
- The contract laboratory is accredited by the state for the methods it runs for the district and reports its quality-control results with each batch of samples.
- District staff review each year's results against field notes, laboratory quality-control reports and earlier years before the report is published.

Data review note (added October 8, 2026)

The first year of Phase 3 results gave the district a new check on its lake data: water leaving a lake through its outlet should carry about the same chloride as the lake itself. That comparison at Penner Lake led to the correction of batch 25-0217 described on page 3. Outlet and lake results are now compared as a standard step in the annual data review.

Data availability

Site-level monitoring data, including the sample results behind Table 3, are available on request under the district's data-request policy at metrolakeswd.org/data-request. Typical turnaround is 10 business days.

Contact

Luis Ferreira (he/him), Communications Coordinator, Metro Lakes Watershed District
(651) 555-0158 | luis.ferreira@metrolakeswd.org

Report history

Date	Edition	Notes
June 11, 2026	First edition	Posted with the district's news release of the same date.
Oct. 8, 2026	Revised edition	Corrects 2025 chloride results for COR-1, PEN-1 and ODG-1 (contract-lab batch 25-0217). Errata on page 3.

Suggested citation

Metro Lakes Watershed District. 2025 Water Quality Report. Revised edition, October 8, 2026. metrolakeswd.org/reports/2025-water-quality.

About the district. The Metro Lakes Watershed District is a local unit of government led by a Board of Managers. It monitors and protects the lakes, ponds and streams of its member cities. Board agendas, budgets and reports are posted at metrolakeswd.org.